

Consumption inequalities in material use undermining resources sustainability

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Growing global material consumption has pushed the Earth beyond safe planetary boundaries, yet little is known about how this burden is distributed across individuals. This study quantifies the inequality in household material footprints (MFs) using detailed global expenditure data from 168 countries, linked with a multiregional input–output approach. Results reveal stark inequalities: the top 10% consumers contribute 36% of global total household MFs, while the bottom 50% of consumers account for merely 18%. Inequality is especially pronounced in non-renewable resources such as metals and fossil fuels. Furthermore, elasticity analysis reveals a recoupling of resource use with high consumption, challenging the notion of absolute decoupling. These findings suggest that current sustainability policies about resource focusing on national efficiency gains may fall short without addressing the material-intensive lifestyles of the affluent. Targeting overconsumption at the top could reduce ecological overshoot and create space for sustainable development and material sufficiency for the global majority.

Human pressures on the Earth system have reached unprecedented levels, pushing multiple planetary boundaries beyond safe limits^{1–3}. This ecological overshoot is largely fuelled by the accelerating global demand for materials^{4–6}, including material extraction, production, consumption and waste disposal, that underpin modern economic activities and lifestyles⁷. In response, material use has emerged as a key indicator of sustainability, leading to the adoption of the material footprint (MF) as a core metric for tracking progress towards sustainable development goals (SDGs) 8.4 and 12.2 (ref. 8). The MF captures the direct and indirect use of materials throughout global supply chains and serves as a measure of resource intensity associated with final consumption^{8,9}. SDG 8.4 and SDG 12.2 specifically target the decoupling of economic growth from environmental degradation. Yet, global material consumption surpassed 100 billion tons in 2021, far exceeding

thresholds considered compatible with a sustainable and equitable global economy¹⁰.

While aggregate statistics highlight the scale of the problem, they often conceal the unequal contributions to material use between and within countries^{7,8,11,12}. Previous studies have documented how material use closely follows patterns of affluence, with higher national gross domestic product (GDP) levels typically associated with larger MFs^{8,9,13–15}. Nonetheless, such national averages obscure important intracountry disparities and ignore how individual- or household-level consumption contributes to resource use and ecological impacts^{16,17}. Economic inequality translates into environmental inequality^{17,18}, where the lifestyles of the affluent impose a disproportionate ecological burden, while vast segments of the population remain materially deprived.

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Existing literature has shown a growing awareness of distributional aspects of environmental footprints^{17,19}, particularly in relation to carbon and energy use^{20–22}. Numerous studies have revealed that higher-income individuals and households consume more energy and generate greater greenhouse gas emissions, both within and across countries^{16,21,23}. However, there remains a critical gap in understanding the distribution of material use at the individual level. While the national and sectoral drivers of MFs are well-studied, relatively little is known about how different consumption patterns across income groups, geographies and lifestyle types, contribute to global material use and overshoot^{24–26}.

Addressing this knowledge gap is vital for two reasons. First, it enables a more accurate assessment of environmental responsibility and the fairness of current consumption patterns. Second, it provides an empirical foundation for designing more effective and equitable policies that align with ecological limits. As many scholars have argued^{17,27,28}, focusing solely on technological solutions and national decoupling targets may be insufficient unless the underlying demand-side drivers of material use, especially excessive consumption by the affluent, are directly addressed.

This study aims to offer the most comprehensive analysis to date of global MF inequality across individuals and consumer groups and by consumption-goods categories. We link an exceptionally detailed global expenditure database, covering 168 countries and 98% of the world's population, to an environmental extended multiregional input–output (EEMRIO) that captures the resource use embodied in consumption. We then quantify inequality in household MFs, disaggregated by material type (biomass, fossil fuels, metals and non-metallic minerals), consumption-goods categories and regions, and calculate MF-Gini coefficients to assess their distributions. The findings reveal stark disparities: the global top 10% of consumers account for 36% of all household material use, while the bottom 50% are responsible for just 18%. This inequality is most severe for non-renewable resources such as metals and fossil fuels, which are predominantly consumed through the purchase of manufactured goods, transportation and energy services. In contrast, biomass use is more evenly distributed, although it still reflects pronounced consumption differences between population groups. These results align with previous research on carbon and energy inequality, but extend the analysis to encompass the full material economy.

We further examine the elasticity of material use with respect to consumption expenditure. While modest relative decoupling is observed at lower income levels, where rising incomes are not matched by equivalent increases in material use, this trend reverses at higher consumption levels. A pattern of recoupling emerges, where additional affluence drives disproportionately high material consumption, particularly of non-renewable resources. This suggests that absolute decoupling of material uses from economic growth is unlikely in the absence of structural changes to consumption patterns and economic systems.

These findings underscore that reducing global material use requires more than efficiency gains or national decoupling targets, it demands confronting the disproportionate consumption of the most affluent. The top 10% far exceed material levels needed for a decent standard of living, while over 70% of the global population remains below that threshold²⁷. Addressing this stark inequality is both an environmental imperative and a matter of social justice. Our study contributes by mapping the global distribution of material use and identifying priority areas for more equitable and sustainable consumption.

Results

In 2017, household consumption accounted for 43% of the global MF (Supplementary Fig. 1), including 74% of biomass use, 56% of fossil fuel use, 39% of metal use and 19% of non-metal use. The remaining 57% of the global MF is attributable to government consumption and capital formation. This portion is not attributed to households in our analysis.

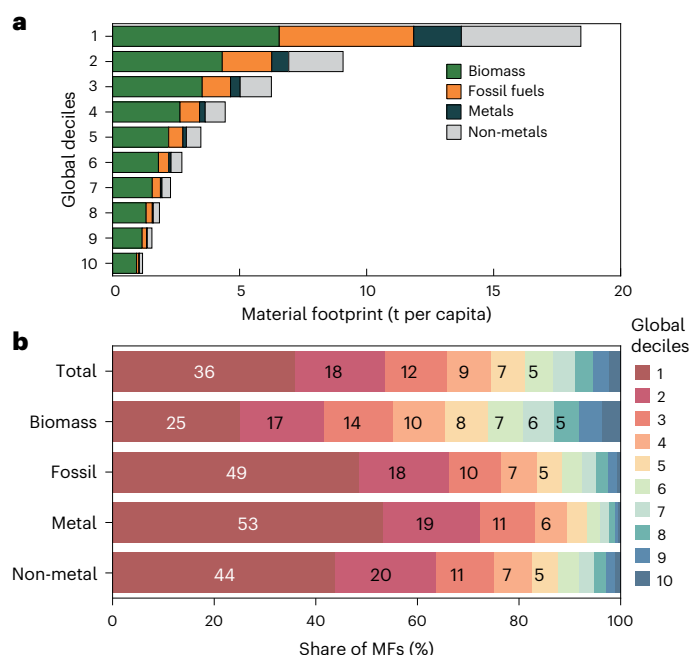


Fig. 1 | MFs by global expenditure decile in 2017. a, Per capita MFs (biomass, fossil fuels, metals and non-metals) across expenditure deciles. **b**, Share of each global expenditure decile in total MFs. Expenditure groups are ordered from highest (decile 1) to lowest (decile 10). The population across the 168 countries is divided into 10 equal-sized global deciles, with each decile representing approximately 0.73 billion people.

Unequal MFs across consumers and consumption categories

Figure 1 presents that the world's top 10% of consumers exert a disproportionate large impact, with a per capita MF of 18.4 t in 2017, 15.6 times higher than that of the bottom 10% (1.2 t per cap) (Fig. 1a). The disparity is even more extreme for metals, where the global top 10% consume 113 times more than the bottom 10%. Compared with the top second decile, the MF of the top first decile is still double, highlighting the outsized role of the top 10% consumers in global material use. Figure 1a and Supplementary Fig. 2 further illustrate that abiotic materials (fossil fuels, metals and non-metals) dominate the MF of high-income groups, whereas biomass makes up a larger share of the MFs among low-income consumers. Nonetheless, wealthier groups have substantially greater MFs across nearly all material types, primarily due to their higher consumption levels.

In terms of total share, the top 10% consumers capture 36% of global MF (Fig. 1b), including 25% of biomass, 49% of fossil fuels, 53% of metals and 44% of non-metals. When expanded to the global top 20%, their share ranges from 42% to 72%, depending on the material type. In stark contrast, the global bottom 10% account for only 2% of the total MF: just 3.6% of biomass, 0.8% of fossil fuels, 0.5% of metals and 2.3% of non-metals. The bottom 50% of the global population use less than 20% of total MFs—only half the contribution of the top 10%. For metals, the inequality is particularly striking: the global top 10% alone account for more than 50% of global total MF, while the bottom 50% contribute just 6.5%.

Figure 2 details the breakdown of MFs by consumption category across global expenditure deciles. It shows that biomass use is largely driven by food consumption across all groups. Fossil fuel use, on the other hand, is mainly linked to food, energy and services, particularly among the affluent. For metals and non-metals, use is primarily contributed by the consumption of manufactured products and services by wealthier groups. It is worth reminding that despite the notable difference in consumption structure of materials, the amounts consumed by the wealthier groups tend to be much greater in almost all consumption categories, thus using more materials compared with those by the poorer groups.

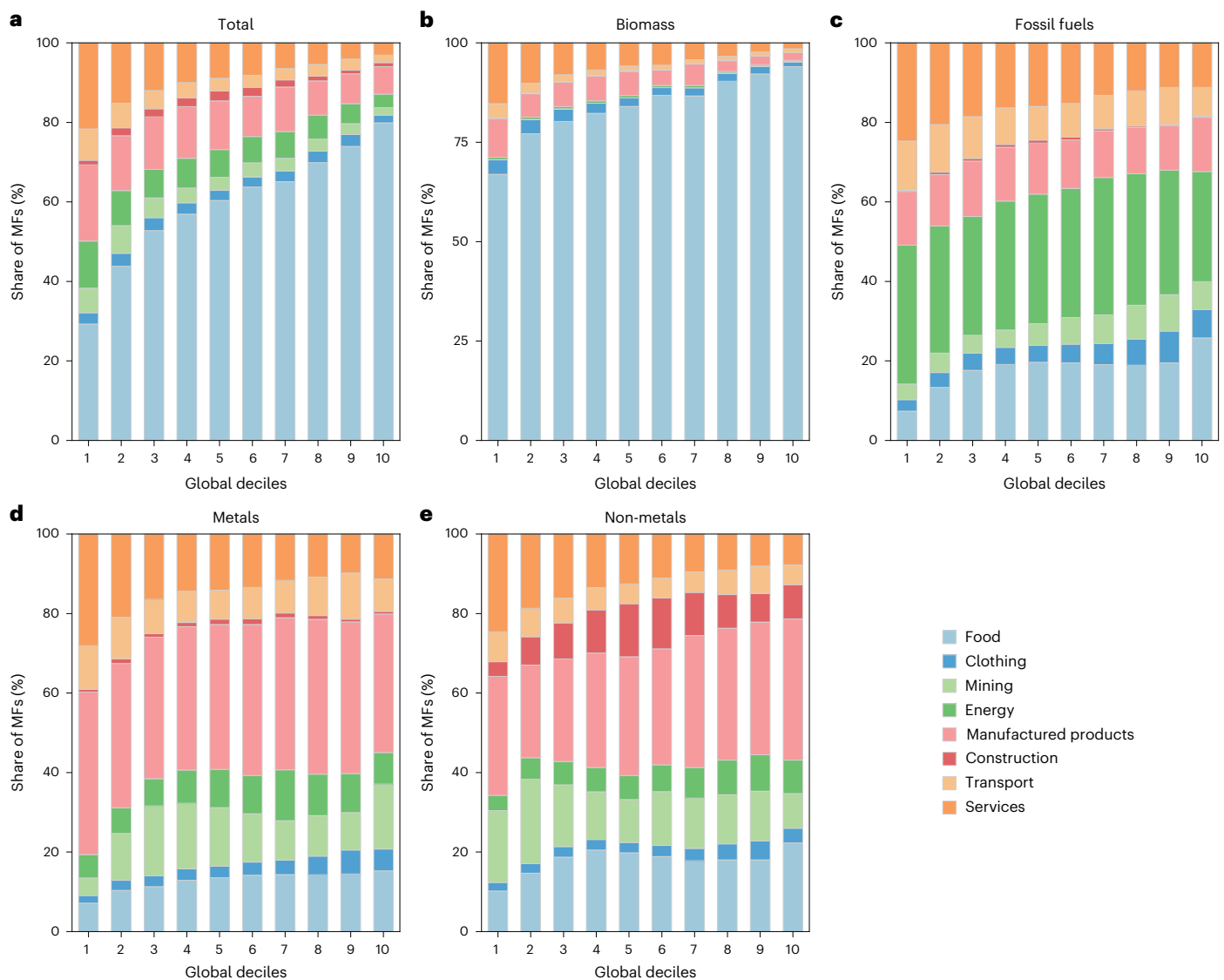


Fig. 2 | Sectoral composition of the MFs across global consumer deciles. Expenditure groups are ordered from highest (decile 1) to lowest (decile 10). **a–e** are sectoral composition of the total (**a**), biomass (**b**), fossil fuels (**c**), metals (**d**) and non-metallic materials (**e**), respectively.

Figure 3a depicts the MF inequality using Lorenz curves and the corresponding Gini coefficients. The overall global MF-Gini is 0.46, indicating a moderate level of inequality. However, disparities are much greater for abiotic materials: metals (0.67), fossil fuels (0.60) and non-metals (0.57), compared with biotic materials such as biomass (0.33). This highlights that abiotic materials are distributed far more unequally across the global population. Notably, the inequality in metal and fossil fuel use exceeds that of other environmental indicators, including carbon emissions and biodiversity loss¹⁷. Despite these disparities, all MF-Gini values remain below the inequality level of global expenditure (0.69) (Supplementary Fig. 3).

Figure 3b disaggregates MF-Gini coefficients by consumption categories. It shows that services (0.66) and transport (0.63) exhibit the highest inequality, while food (0.30) and construction (0.43) are more equally distributed. Inequality also varies by material type within consumption-goods categories. For example, the MF-Gini for metals in services, transport and manufactured goods reaches 0.75, 0.70 and 0.68, respectively, while the biomass MF-Gini for food is just 0.28. These findings suggest that MF inequality is primarily driven by abiotic materials consumption in the secondary and tertiary sectors.

Unequal MFs across and within countries

Figure 4 illustrates the per capita MFs of household consumption across 168 countries in 2017, revealing substantial variation between countries. Average per capita MFs are strongly correlated to the national expenditure (income) level: developed countries generally exhibit much higher footprints. For instance, Canada and Switzerland both exceed 20 t per capita, more than four times the global average; while low-income countries such as the Democratic Republic of Congo, Burundi and Lesotho report average MFs below 2 t per capita. Despite representing over one-third of the global population, China and India both fall below the global average in per capita MF. However, exceptions exist: Mongolia (12.9 t per capita), for example, shows high per capita MF despite being a developing country, highlighting the material intensity (MF per unit of final consumption) of certain lifestyles. Interestingly, global north countries, especially those in Europe and North America, tend to exhibit lower material intensity per unit of consumption, whereas global south countries generally show higher material intensity (Supplementary Fig. 4). This pattern arises because high-income countries typically allocate larger share of their expenditure to sectors with lower material intensity, such as services, while low-income countries spend a greater share on essential goods such

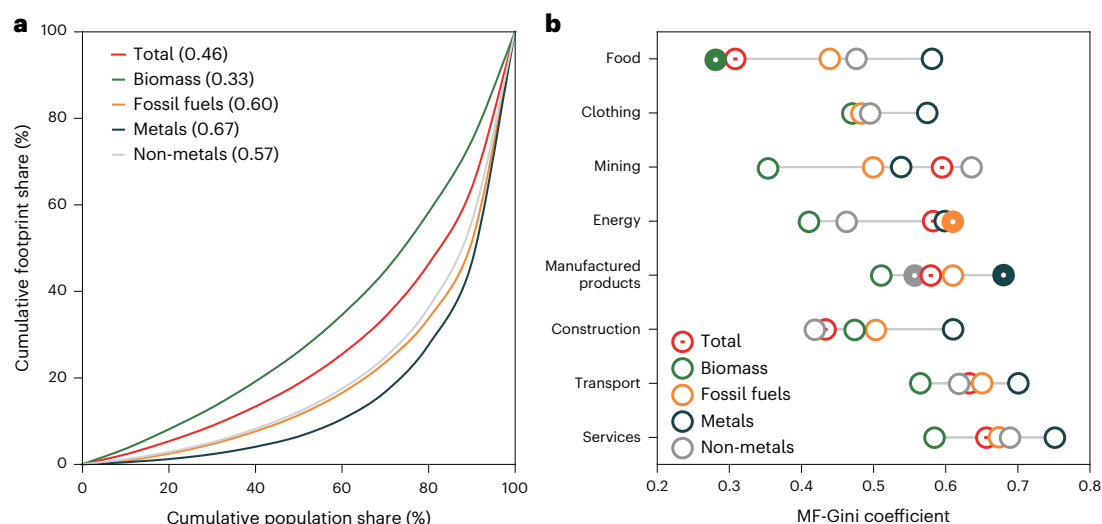


Fig. 3 | MF inequality in 2017, measured by Lorenz curves and Gini coefficients. a, Lorenz curves of MFs on the global scale. **b**, MF-Gini coefficients by eight consumption-goods categories. Thicker circles denote the primary sectors of material use for each of the four material types (biomass, fossil fuels, metals and non-metallic minerals).

as food and energy, which are inherently more resource-intensive. Nevertheless, it is important to emphasize that affluent countries still maintain markedly higher total MFs, driven by their substantially greater levels of absolute consumption.

To examine intraregional disparities, countries were grouped into six major regions, with China, India and the USA analysed separately (Fig. 4b). Notable inequality is observed within regions. The national top 10% in the USA exhibit the highest per capita MF globally at 76.9 t per capita, nearly triple the national second decile (28.1 t per capita). Similarly, the regional top decile in developed Asia-Pacific countries reaches 31.2 t per capita. In contrast, even the national top 10% of consumers in India have lower MFs than the bottom segments of developed regions. Notably, the top 10% in China and the Latin America and the Caribbean have per capita MFs exceeding those of most individuals in Europe and the Asia-Pacific developed regions. This indicates that elite consumers in developing regions also substantially contribute to global material demand.

Regional inequality is further assessed using the MF-Gini coefficient (Supplementary Fig. 5). Our results show that intraregional inequalities in material use are generally smaller than the global-level inequality. Among the regions analysed, the USA exhibits the highest regional MF-Gini at 0.45, followed by China (0.39). In contrast, Europe has the lowest regional MF-Gini at 0.20, reflecting its stronger efforts in alleviating inequality^{29,30}. The results are also consistent with the previous studies on income and environmental inequality^{20,21}, reinforcing the conclusion that economic development and inequality alleviation can be pursued simultaneously.

This pattern is further confirmed at the national level, as shown in Supplementary Fig. 6, which presents MF-Gini coefficients for all 168 countries analysed. Most European countries have relatively low inequality, whereas three African countries, Malawi, Namibia and Rwanda, have the highest MF-Gini. Some high-income countries such as the USA and Canada also report relatively high levels of MF inequality. Correlation analysis (Supplementary Fig. 6) suggests that higher national average expenditures are generally associated with lower MF-Gini coefficients, except for the USA. Finally, inequality patterns differ by material type. In most countries, biomass use is the most equally distributed, while metal use shows the highest inequality, which is in line with the global pattern.

Responsibility for material overshoot

The unequal MF distribution translated into starkly differentiated responsibilities for exceeding planetary boundaries. To better

understand this, we examine the ratio of per capita MF to the per capita fair share of household consumption (Methods). As shown in Fig. 5a, the MFs of the top global consumers far exceed sustainable threshold for household consumption (approximately 3.1 t per capita in 2017). Specifically, the top 10% and the second decile surpass this boundary by approximately 610% and 300%, respectively. On the contrary, the global bottom 50% remains within proposed material use thresholds. Disaggregating by material type, we observe that top consumers disproportionately overshoot abiotic material boundaries, while their overshoot of biotic materials is comparatively lower. Conversely, among lower-income consumers, especially those in the bottom deciles, the MF-to-limit ratio is relatively higher for biotic materials than for abiotic ones.

Figure 5b highlights the disproportionate contributions of global expenditure deciles to total material overshoot. The global top decile of consumers are responsible for 57%, 41% and 67% of the overshoot in total, biotic and abiotic materials use, respectively. The global second decile contributes 23% of the total overshoot. By contrast, the bottom 50% of the population contributes negligibly to global material overshoot. These findings underscore that the top decile of consumers is the main contributor to material overshoot, especially in relation to non-renewable (abiotic) materials. It is worth noting, however, that this attribution of responsibility is confined to household consumption and does not encompass MFs associated with capital formation, which may shift the distributional dynamics in some contexts.

The material elasticity of consumption

Previous studies have examined the coupling/decoupling between material use and economic growth using national-level MF time series^{7,15}. In contrast, our study investigates the global MF elasticity of consumption (MF elasticity), a metric that quantifies the proportional response of material use to changes in household consumption, based on high-resolution consumption segments data.

Our results show that MF elasticity varies substantially by material type (Supplementary Table 1). The elasticity of the biomass footprint is about 0.459, while that for non-renewable abiotic materials is markedly higher, ranging from 0.836 to 1.102. This indicates that increases in consumption levels are associated with disproportionately higher demand for non-renewable resources. Notably, metal footprints exhibit the highest elasticity (1.102), suggesting that a 1% increase in household expenditure results in a roughly equivalent 1.102% rise in metal use.

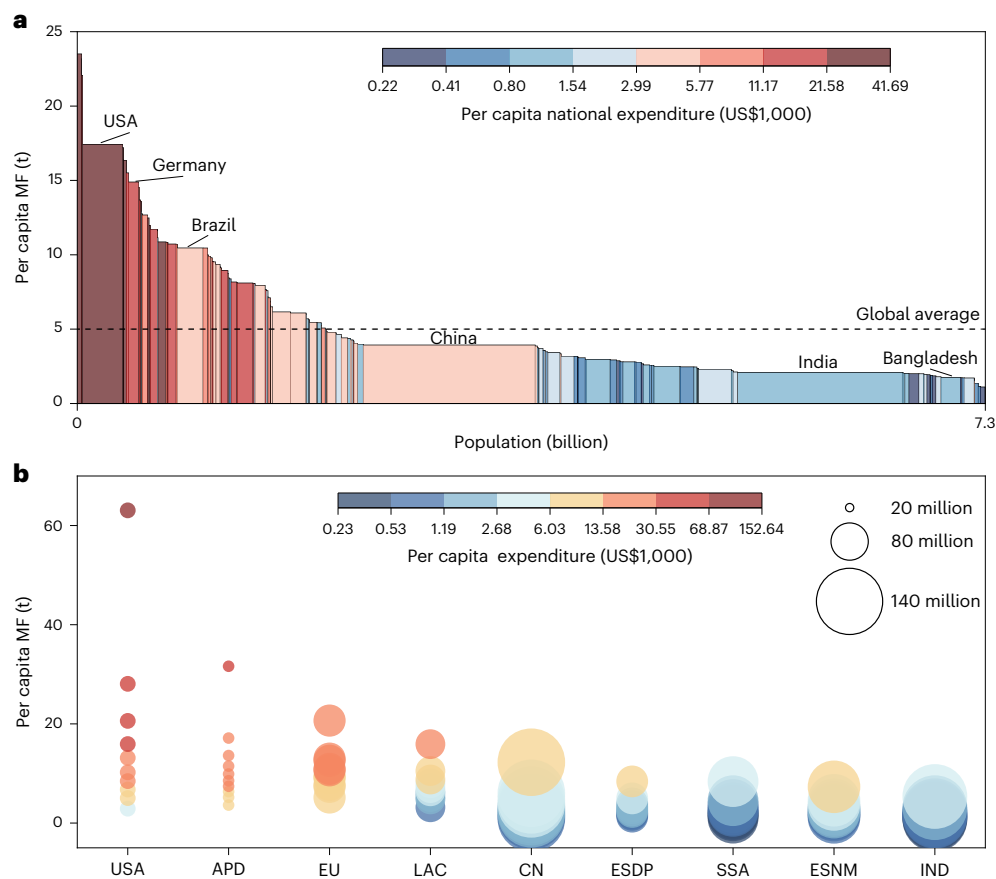


Fig. 4 | Per capita MFs of household consumption by country and expenditure decile. **a**, Per capita MFs of household consumption and population sizes for 168 countries. **b**, Per capita MFs and population sizes by regional/national expenditure decile. Bubble size represents the population size of the corresponding expenditure decile in each region/country. Regional

abbreviations: EU, Europe; CN, China; APD, Asia-Pacific developed; ESDP, East and Southeast Asia and developing Pacific; LAC, Latin America and the Caribbean; IND, India; SSA, sub-Saharan Africa; and ESNM, Eurasia, Southern Asia, North Africa and the Middle East.

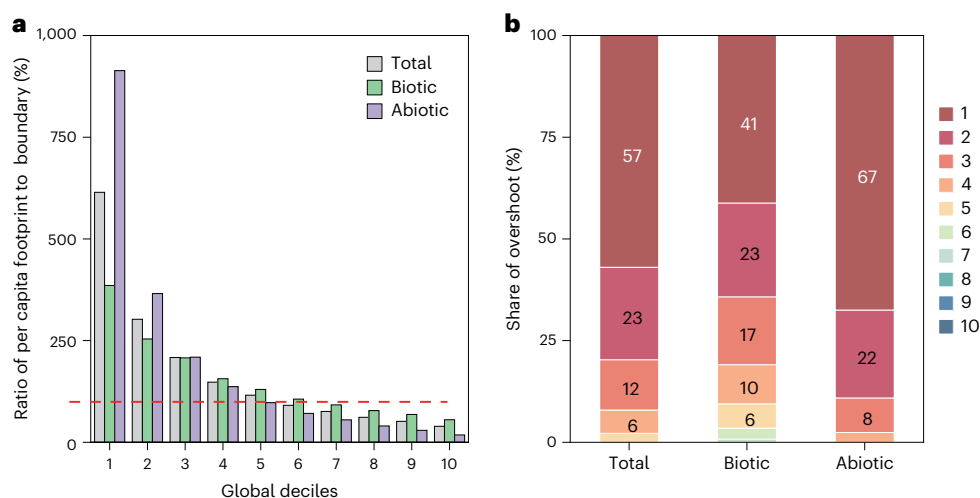


Fig. 5 | Contribution of global expenditure deciles to material overshoot. **a**, Ratio of per capita MF to the per capita material sustainable level for household consumption. Deciles are ordered from the highest (group 1) to the lowest (group 10) in terms of per capita expenditure. **b**, Share of global material overshoot

attributable to each expenditure decile. Overshoot is assessed relative to proposed sustainable boundaries for total, biotic and abiotic material use. Legend labels 1–10 in **b** denote expenditure deciles from highest (1) to lowest (10).

However, MF elasticity is not uniform across expenditure (income) groups. As illustrated in Fig. 6, elasticity values across global expenditure deciles show a nonlinear pattern. For all four material types, elasticity tends to be higher, or shows an increasing trend, among the

lower global deciles, particularly for metals and non-metals, probably reflecting the material intensity of poverty alleviation and basic infrastructure development. This is followed by a marked decrease across the middle and upper-middle deciles. Yet, at the highest expenditure

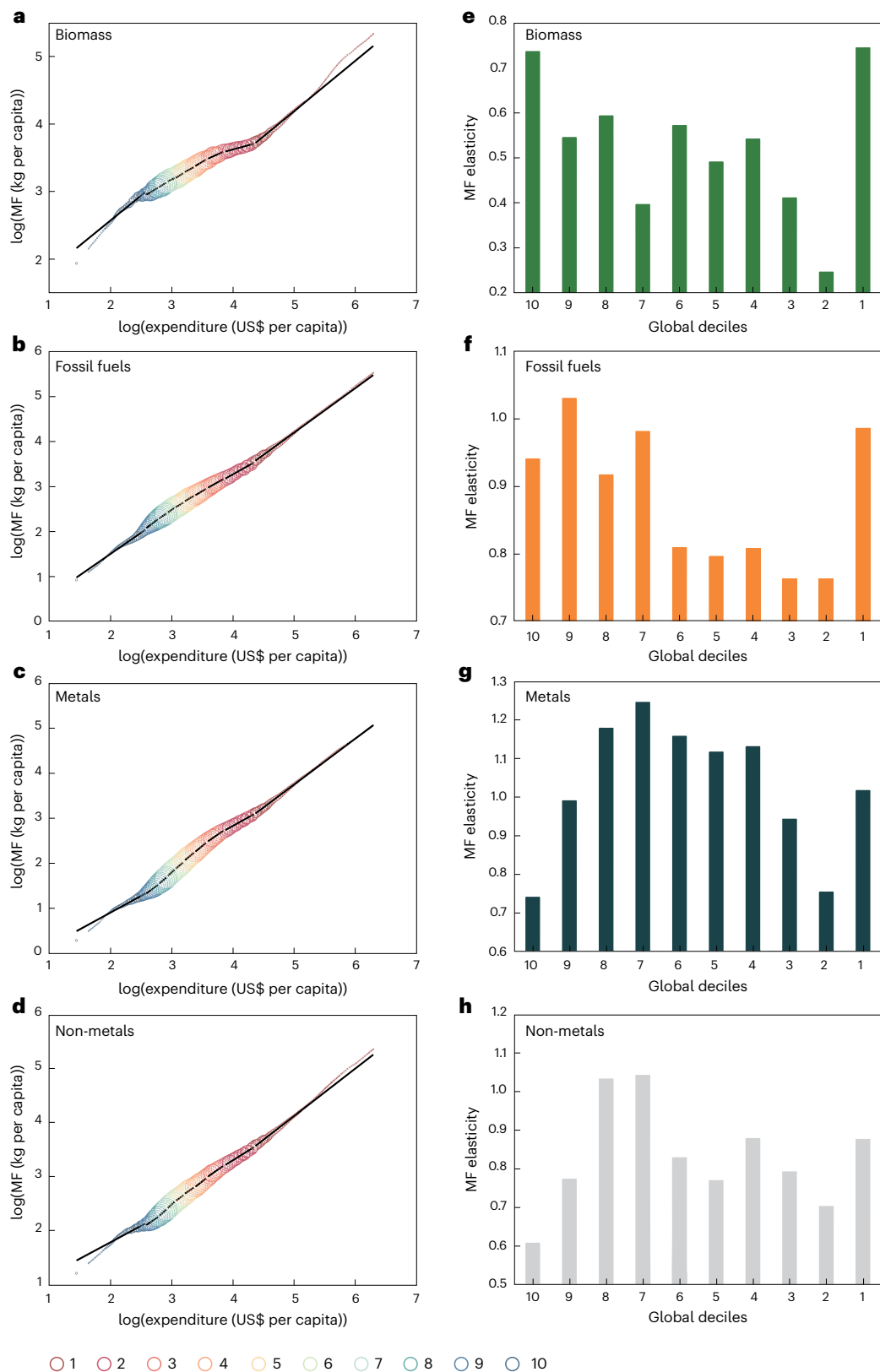


Fig. 6 | The MF elasticity of consumption for four material types. a–d, The log–log relationships between per capita expenditure and per capita MF for biomass (a), fossil fuels (b), metals (c) and non-metallic minerals (d); circles denote expenditure deciles from highest (1) to lowest (10) and are scaled by population

size. **e–h,** Variations in MF elasticity across global expenditure deciles (from highest to lowest: group 1 to group 10). From top to bottom, they correspond to biomass (e), fossil fuels (f), metals (g) and non-metallic minerals (h). All elasticity estimates are statistically significant ($P < 0.001$) based on two-sided t -tests.

levels, all MF-elasticities rise sharply again. This nonlinear pattern of changes reflects a process of relative decoupling at intermediate consumption levels, followed by recoupling as consumption becomes excessive. These findings suggest that widespread absolute decoupling of material use from consumption growth remains unlikely under current technological and structural conditions. In other words, continued pursuit of ever-higher living standards, particularly among the affluent, risks reinforcing material-intensive consumption patterns.

Further supporting this pattern, Supplementary Fig. 8 presents national-level MF elasticity estimates. Across most countries, material use increases strongly with rising consumption. For example, in 103 of the 168 countries analysed, including both India and the UK, the elasticity of metal footprints exceeds 1, indicating no evidence of decoupling between metal use and household consumption expenditure.

Discussion

Devising equitable and sustainable solutions within the limits of the ecological capacity of the Earth necessitates a detailed understanding of how material use is distributed across populations and how this relates to patterns of consumption. While previous studies have examined MFs primarily at the national level^{8,9,13,31}, our study provides a much more granular picture by assessing the distribution of household MFs across global, regional, national and consumption-goods category levels. This approach not only confirms existing findings on cross-country inequality^{15,18} but also reveals the substantial disparities within countries and across consumer groups, which are often masked by national averages.

Our results show that global material use associated with household consumption is highly skewed: the global top 10% of consumers are responsible for 36% of household MFs, while the bottom 50% contribute less than 20%. This inequality becomes even more pronounced when examining responsibility for material overshoot. The top 10% alone contribute 57% of global overshoot, including 67% of overshoot related to non-renewable abiotic materials such as fossil fuels and metals. The global top 20% is responsible for nearly 80% of the overshoot. In contrast, the bottom 50% of the population contribute minimally to this overshoot and largely remain within sustainable material use thresholds. Such findings clearly indicate that excessive consumption by affluent groups, rather than universal overuse, is the primary driver of ecological overshoot.

This unequal distribution reflects a broader trend: rising affluence is strongly coupled with increasing material use, particularly non-renewable resources. Although global sustainability efforts have largely focused on improving production-side efficiency and decoupling material use from GDP growth at the national level^{8,9,32,33}, our study suggests that such strategies may be insufficient. We observe substantial disparities in MFs even within countries, where differences between high- and low-income consumers often exceed differences across nations. These within-country inequalities are often overlooked in global sustainability policy discourse but are increasingly relevant given the widening income gaps observed globally^{16,34}.

A notable aspect of our analysis is the estimation of MF elasticity with respect to expenditure. This metric reveals how sensitive material use is to changes in household consumption. Our results show that the MF elasticity of biomass use is relatively modest (~0.45), while that of non-renewable abiotic materials is much higher, ranging from 0.82 to 1.02. Notably, metals exhibit the highest elasticity, indicating that increases in income translate almost proportionally into greater material use. This underscores the high material intensity of affluent lifestyles, especially in relation to infrastructure, mobility and luxury goods.

Moreover, the MF elasticity of material use is not constant across expenditure levels. We find that MF elasticity exhibits a nonlinear pattern across global deciles. Elasticities tend to increase rapidly at the lowest income levels, probably reflecting the resource needs associated with basic services and infrastructure during early stages of development. They then decline across the middle-income groups, suggesting some

relative decoupling. However, among the wealthiest deciles, elasticities rise again sharply, reflecting a recoupling of consumption and material use. This pattern indicates that while material use may stabilize or grow slowly at intermediate levels of consumption, it accelerates once affluence reaches a certain threshold. Such a recoupling process calls into question the feasibility of achieving absolute decoupling under current technological and economic structures, especially when the consumption ambitions of affluent groups remain unchecked.

Our findings have important implications for sustainability policy. Current approaches, which largely rely on technological innovation and efficiency improvements³⁵, must be complemented with demand-side measures that address overconsumption^{17,36}. In particular, targeted policies addressing the consumption patterns of the most affluent groups, who are disproportionately responsible for material use and ecological overshoot, are essential. Such measures may include progressive resource taxation, regulation of luxury consumption and incentives to promote low-impact lifestyles³⁷. In addition, industry and business sectors can play a crucial guideline role by transforming material use and packaging systems, committing to zero-waste strategies and embracing circular economy principles³⁸. At the same time, ensuring access to a minimum material standard of living for the majority remains a key priority²⁸. Our study finds that the top 10% have an average MF of 18.4 t per capita, more than three times the estimated threshold (~6 t per capita) required for a decent standard of living²⁷. In contrast, more than 70% of the global population falls below this threshold. Thus, reallocating material use through equitable policies could not only reduce global overshoot but also improve material well-being for a large share of the population.

Finally, the composition of material use also varies across consumption-goods categories. Our results indicate that the affluent disproportionately consume non-renewable abiotic materials, particularly metals, because of their reliance on industrial goods, personal vehicles and resource-intensive services. Meanwhile, biomass use, although more evenly distributed, still constitutes a substantial share of total material use, particularly in low-income regions. Inefficiencies in biomass use, such as those associated with meat-heavy diets³⁹, further highlight the need for more sustainable consumption practices across all income levels^{27,40}.

In sum, this study demonstrates that material inequality is a fundamental barrier to sustainable development. The current trajectory of rising consumption among the global affluent not only undermines environmental limits but also restricts opportunities for the majority to attain a decent quality of life. Addressing this challenge requires a paradigm shift in sustainability discourse, from a focus on aggregate efficiency to one that centres equity, sufficiency and differentiated responsibilities. Only through such a shift can we hope to achieve sustainability for resources use within the ecological boundaries of the Earth.

Methods

The EEMRIO model. The EEMRIO approach is a classic method for calculating environmental, economic and societal footprints^{17,41–43}. One of its key advantage is its ability to capture both direct and indirect environmental impacts along the entire supply chains of final consumption-goods and services^{8,13,17}. At its core, the MRIO table captures the monetary transactions and intersectoral relationships across industries (*j*) and regions or countries (*i*). By combining environmental satellite accounts with the MRIO framework, we quantify the MFs embodied in final consumption. The basic structure of the model is described by the Leontief demand model:

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1} \mathbf{f} \quad (1)$$

where $\mathbf{x}$ refer to the total output vector ($ij \times 1$), $\mathbf{I}$ the identity matrix, $\mathbf{A}$ the technical coefficients matrix, $(\mathbf{I} - \mathbf{A})^{-1}$ the Leontief inverse matrix ($ij \times ij$) and $\mathbf{f}$ the final demand vector by sector ($ij \times 1$).

The MF of household consumption for a given material type c is calculated by the EEMRIO model as:

$$ef_{i,q,c} = E_c(I - A)^{-1}y_{i,q} \quad (2)$$

where $ef_{i,q,c}$ is the MF of consumer group (q) in country/region (i) for material type (c). Parameter $y_{i,q}$ is the final demand vector of consumer group (q) in the country/region (i); E_c is the material intensity diagonal matrix for material type c , which links sectoral output to resource use. The elements of the material intensity matrix E_c are defined as:

$$E_c = \text{diag}[e_{i,j,c}], \text{ where } e_{i,j,c} = k_{i,j,c}/x_{i,j} \quad (3)$$

In equation (3), $e_{i,j,c}$ represents the material use intensity of sector j in country i for material type c , $k_{i,j,c}$ is the total resource use by sector j in country i and $x_{i,j}$ is the total output of that sector.

The consumer group final demand vector $y_{i,q}$ is derived by bridging household expenditure survey data with the final demand of MRIO table. Details of this bridging process are provided in the section ‘Data source and processing’ (equation (8)), where we explain how expenditure shares from household expenditure survey are matched to sectoral household final consumption categories within the MRIO framework.

Inequality measurements. The Gini coefficient is typically used to measure income and wealth inequalities¹⁷. It ranges from 0 to 1, indicating complete equality to complete inequality^{34,44}. Here we used it to estimate MF inequalities among consumer groups, which can be expressed as:

$$G = \sum_{q=1}^n P_q Y_q + 2 \sum_{q=1}^n P_q (1 - C_q) - 1 \quad (4)$$

where G refers to the MF-Gini coefficient, P_q , Y_q and C_q represent population share, MF share and the cumulative MF share of consumption group q , respectively. Parameter n stands for the number of groups.

We also illustrate MF inequality with a Lorenz curve, which charts cumulative MF share against accumulative population share sorted by MF.

Quantifying the responsibility for overshoot. In this study, the method developed by ref. 15 is used to estimate the responsibility for material overshoot. The ‘sustainable level’ of material use is based on proposals by industrial ecologists, with various upper limits suggested in the literature^{45–47}. For example, ref. 46 proposed a global extraction cap of 50 Gt yr^{−1} and a per capita target of 8 t yr^{−1} by 2030, while the International Resource Panel of UNEP⁴⁵ recommended 6–8 t per year per capita by 2050. Others⁴⁷ proposed a more stringent threshold of 5 t per year per capita. We acknowledge that determining a sustainable level of material use remains a complex and evolving field of research, involving both scientific uncertainties and normative judgements. In this study, we adopt a global budget of 50 Gt yr^{−1}, which is widely used in the literature as a reference point^{15,43}. Following convention, we allocate one-third of this total to biotic materials and two-thirds to abiotic materials^{15,43,47}. Using an equal per capita allocation method^{15,18,48}, this translates into a per capita material use threshold of about 7 t yr^{−1} for 2017.

However, household consumption constitutes only a portion of total material use. To account for this, we further refine the per capita threshold to reflect household consumption only. Using final demand for MF in 2017, we identified the share of the total MF attributable to household consumption and applied this fraction to the 7 t per capita global threshold. This calculation results in a per capita material use budget for household consumption of approximately 3.1 t per capita per year (for 2017).

On the basis of the established thresholds, we quantify the responsibility for material overshoot by calculating two key indicators: the

‘exceedance ratio’ and the ‘share of overshoot’¹⁷. First, we compute the exceedance ratio, which reflects the extent to which a consumption group surpasses its fair share of the material use budget. The exceedance ratio for consumption group q is defined as:

$$\text{Exceedance ratio}_q = \frac{EF_q - \text{Share}_q}{\text{Share}_q}, \quad (5)$$

in which EF_q denotes the MF of group q and Share_q represents that group’s fair share of the material use budget. The fair share is determined by multiplying the per capita material use budget by the population size of group q .

Second, we calculate the ‘share of overshoot’, which expresses the proportional responsibility of each group for global material overshoot. Following ref. 15, the undershoot of one group does not compensate for the overshoot of another. Accordingly, for any group whose MF falls below its fair share, the exceedance ratio is set to zero in this calculation. The share of overshoot for group q is given by:

$$\text{Share of overshoot}_q = \frac{\text{Exceedance ratio}_q}{\text{Exceedance ratio}_{q,\text{total}}} \quad (6)$$

MF elasticity of expenditure. To estimate the elasticity of MF relative to household spending (MF elasticity), we perform a population-weighted log–log regression of per capita MF (M) on per capita expenditure (E), following the approach used in previous studies^{21,49}. The regression model is specified as:

$$\log M = a + b \log E \quad (7)$$

In equation (7), the coefficient b represents the elasticity of MF with respect to household expenditure. It quantifies the percentage change in per capita MF associated with a 1% change in per capita expenditure⁵⁰. Specifically, a value of $b > 1$ indicates that material use increases more than proportionally with rising expenditure, whereas $b < 1$ implies a less-than-proportional increase.

For estimating the global MF elasticity, we use the 201 standardized expenditure levels provided in the global household expenditure dataset, which are consistent across countries. For each level, the corresponding MF is calculated using the MRIO framework. The elasticity is then estimated on the basis of the relationship between MF and expenditure across these 201 global data points.

For estimating the global decile MF elasticity, the same 201 expenditure levels are grouped into ten global expenditure deciles and the elasticity is calculated separately for each decile. This allows us to capture the nonlinear relationship between expenditure and material use across different global income groups.

For calculating the national MF elasticity, we estimate MF elasticity using the national deciles of household expenditure and their associated MFs. This enables comparison of elasticity values across countries and regions.

Data source and processing. The MRIO table is obtained from the GLORIA MRIO database^{8,51,52} developed by the University of Sydney using the IELab infrastructure. This database covers 164 regions and 120 economic sectors and represents the most recent and detailed MRIO dataset specifically tailored for MF analysis. We use the GLORIA059a version, which covers the year 2017. The corresponding materials satellite accounts for 2017, also obtained from the GLORIA database, follow the standard methodology of UN IRP⁵³. These accounts categorize material flows into four major materials groups—biomass, fossil fuels, metal ores and non-metallic minerals—further disaggregated into 62 detailed subcategories (23 for biomass, 10 for fossil fuels, 15 for metal ores and 14 for non-metallic minerals), in accordance with the common compilation categories used by the UN IRP⁵³. For more details on the

construction of the MRIO tables and materials extensions, please refer to ref. 8. All four materials groups are included in our MF estimations.

The household expenditure survey data used in this study covers 168 countries and approximately 98% of the population of the world. This composite dataset is primarily based on the World Bank Global Consumption Database (WBGCD), which involves detailed consumption information in 201 expenditure levels. To ensure completeness and coverage, we supplemented the WBGCD with additional data from national household surveys for countries not included in the original database. These supplementary sources include the Eurostat household budget survey, the Japanese family income and expenditure survey, the Canada survey of household spending and the Australia household expenditure survey¹⁷. Further details on the construction and harmonization of this dataset are provided in refs. 17,20.

The household expenditure data are systematically bridged and matched to the GLORIA MRIO framework to estimate the MFs of different household consumption groups. Specifically, we construct a bridging matrix to map national household expenditure profile to the household final consumption categories in the GLORIA MRIO table. In this matching process, the household consumption structure in GLORIA serves as the benchmark, ensuring consistency with the underlying MRIO sectoral classification. This bridge matrix is constructed individually for every country, enabling the derivation of relative consumption shares across sectors for each expenditure group. These shares are aligned with the GLORIA sectoral classifications, allowing for detailed, group-level disaggregation of household demand. A schematic illustration of this disaggregation process is provided in Supplementary Fig. 9. The matching process is mathematically expressed as:

$$\mathbf{y}_{i,q} = \mathbf{b}_{i,q} \mathbf{f}_i \quad (8)$$

where $\mathbf{f}_i$ is the initial total household demand vector ($ij \times 1$) for country/region (i) in MRIO table; $\mathbf{y}_{i,q}$ is the final consumption vector ($ij \times 1$) of consumer group (q) in the country (i); $\mathbf{b}_{i,q}$ is bridge matrix, derived from household survey data. The elements of the bridging matrix are calculated as:

$$\mathbf{b}_{i,q} = [\mathbf{b}_{i,j,q}], \text{ where } \mathbf{b}_{i,j,q} = \text{Expenditure}_{i,j,q} / \text{Expenditure}_{i,j,\text{total}} \quad (9)$$

here $\text{Expenditure}_{i,j,q}$ is the expenditure of consumption group (q) in country/region (i) for sector (j) and $\text{Expenditure}_{i,j,\text{total}}$ is the total expenditure in country/region (i) for sector (j).

Notably, this integration is conducted in basic prices (that is, producer prices), ensuring consistency with the pricing convention of the MRIO table and avoiding distortions due to purchaser price difference across countries in the household survey data. It is worth noting that the matched data represent relative expenditure shares across consumption groups rather than absolute monetary values, consistent with established methods in the literature^{17,54}. This process yields a harmonized, high-resolution dataset of household final demand, disaggregated by both consumption-goods categories and income-based consumer groups, fully compatible with the GLORIA MRIO structure and suitable for environmental footprint analysis across consumer groups.

Limitation. Several limitations and sources of uncertainties should be acknowledged in this study. First, our estimation of household consumption MFs relies on an EEMRIO model. The primary uncertainties in the EEMRIO approach stem from the underlying MRIO tables and the environmental extension accounts on which they are built^{55–57}. Key sources of uncertainty, ranked from greatest to least, include: the accuracy of material extension accounts, the allocation of material use across economic sectors, the resolution and structure of final household consumption data and the overall configuration of the economy. Furthermore, because of the linear structure of the Leontief production

function, EEMRIO models cannot differentiate between goods of varying quality or price, such as a basic versus a luxury vehicle, potentially underestimating the environmental impacts associated with high-end consumption patterns⁵⁸. Despite these limitations, the EEMRIO framework remains one of the most widely used tools for consumption-based environmental impacts accounting. Second, survey-based expenditure data are prone to under-reporting and sample bias. Respondents may misremember or omit infrequent purchases, contributing to recall bias⁵⁹. Moreover, certain demographic groups, particularly the ultra-wealthy, are often under-represented in household surveys. As a result, the MF and associated inequality of the wealthiest individuals may be underestimated⁶⁰. Third, this study focuses exclusively on the MFs associated with global household consumption. It does not include material requirements linked to other final demand categories, most notably capital formation. Previous studies have shown that capital investments, particularly in developing countries, are a major driver of rising material use^{8,61}. Moreover, the MF of household consumption in our study reflects only the inputs required in the reference year (2017). It does not capture legacy material inputs from capital goods accumulated in previous years, as MRIO tables account only for intermediate consumption and do not include embedded capital stocks such as machinery, infrastructure and buildings⁶². Therefore, to achieve a more comprehensive understanding of the inequality and responsibility of material use, future research should seek to extend the analysis to include capital formation and capital stock flows, ideally at subnational and sector-specific levels. This represents an important and promising direction for advancing sustainability footprint accounting.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The GLORIA database, including the MRIO table and materials satellite accounts, is openly accessible at <https://ielab.info/>. Expenditure data are collected from the World Bank (<https://data.worldbank.org/>) and other sources. See ref. 17 or contact the corresponding authors for more details.

Code availability

No new code is developed in this study.

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Author contributions

P.T., K.F., D.L. and L.S. designed the study. P.T. and D.L. performed the analysis and prepared the paper. K.F., D.L. and L.S. coordinated and

supervised the project. P.T., K.F., X.C., D.L., M.J., J.L., H.Z., Y.S. and L.S. participated in writing the paper.

Competing interests

The authors declare no competing interests.

Additional information

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Study description

Global material consumption has grown rapidly, pushing the Earth system beyond safe planetary boundaries. However, the distribution of materials used across individuals remains poorly understood. In this study, we quantify inequality in household material footprints (MFs) by linking high-resolution global expenditure data from 168 countries, representing 98% of the world's population, with an environmentally extended multi-regional input-output (EEMRIO) model. Our findings reveal striking disparities: the top 10% of global consumers are responsible for 36% of all household material use, while the bottom 50% account for just 18%. Inequality is particularly acute for non-renewable resources such as metals and fossil fuels, which are embedded in manufactured goods, transportation, and energy services. These results extend the study of environmental inequality beyond carbon emissions to the broader material economy.

We further examine the elasticity of material use relative to household expenditure. While modest relative decoupling is observed at lower income levels, where increases in consumption are not matched by equivalent increases in material use, this trend reverses among the most affluent. A pattern of re-coupling emerges: at high expenditure levels, material use, particularly of non-renewables, accelerates disproportionately. This challenges the perception that economic growth can be absolutely decoupled from environmental impact under current structure and technologies.

Together, these insights suggest that policies solely focus on national material efficiency may be insufficient. Addressing the material-intensive lifestyles of the affluent is essential for reducing ecological overshoot and enabling material sufficiency and sustainable development for the global majority.

Research sample	The research covers 168 countries, which account for 98% of the global population and 97% of the global GDP. In each country, the population is divided into 201 expenditure bins.
Sampling strategy	We have integrated as much of the available data as possible, including the GLORIA MRIO table, the household expenditure survey. Finally, 168 countries were included in the analysis.
Data collection	The MRIO table used in the paper is derived from the GLORIA MRIO database (https://ielab.info/). For household expenditure data, multi-source household survey data is used, i.e., World Bank Global Consumption Database (WBGCD) is available at the database website (https://datatopics.worldbank.org/consumption/). The Household Budget Survey (HBS) for European countries is obtained from Eurostat (https://ec.europa.eu/eurostat/web/household-budget-surveys). The Family Income and Expenditure Survey (FIES) for Japan is provided by Statistics Bureau of Japan (https://www.stat.go.jp/english/data/sousetai/1.html). The Survey of Household Spending (SHS) for Canada is provided by Statistics Canada (https://www.statcan.gc.ca/en/survey/household/). Household Expenditure Survey (HES) of Australia is provided by Australian Bureau of Statistics (https://www.abs.gov.au/statistics/economy/finance/household-expenditure-survey-australia-summary-results). More details can be found in (https://doi.org/10.1038/s41586-024-08154-w).
Timing	Household data was collected from each database's website in September 2022. The MRIO table was collected in April 2024.
Data exclusions	n/a
Non-participation	n/a
Randomization	n/a

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☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.

Validation

Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)

State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.

Authentication

Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.

Mycoplasma contamination

Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.

Commonly misidentified lines
(See [ICLAC](#) register)

Name any commonly misidentified cell lines used in the study and provide a rationale for their use.

Palaeontology and Archaeology

Specimen provenance

Provide provenance information for specimens and describe permits that were obtained for the work (including the name of the issuing authority, the date of issue, and any identifying information). Permits should encompass collection and, where applicable, export.

Specimen deposition

Indicate where the specimens have been deposited to permit free access by other researchers.

Dating methods

If new dates are provided, describe how they were obtained (e.g. collection, storage, sample pretreatment and measurement), where they were obtained (i.e. lab name), the calibration program and the protocol for quality assurance OR state that no new dates are provided.

☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.

Ethics oversight

Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Animals and other research organisms

Policy information about [studies involving animals](#); ARRIVE [guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals	<i>For laboratory animals, report species, strain and age OR state that the study did not involve laboratory animals.</i>
Wild animals	<i>Provide details on animals observed in or captured in the field; report species and age where possible. Describe how animals were caught and transported and what happened to captive animals after the study (if killed, explain why and describe method; if released, say where and when) OR state that the study did not involve wild animals.</i>
Reporting on sex	<i>Indicate if findings apply to only one sex; describe whether sex was considered in study design, methods used for assigning sex. Provide data disaggregated for sex where this information has been collected in the source data as appropriate; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex-based analyses where performed, justify reasons for lack of sex-based analysis.</i>
Field-collected samples	<i>For laboratory work with field-collected samples, describe all relevant parameters such as housing, maintenance, temperature, photoperiod and end-of-experiment protocol OR state that the study did not involve samples collected from the field.</i>
Ethics oversight	<i>Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.</i>

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration	<i>Provide the trial registration number from ClinicalTrials.gov or an equivalent agency.</i>
Study protocol	<i>Note where the full trial protocol can be accessed OR if not available, explain why.</i>
Data collection	<i>Describe the settings and locales of data collection, noting the time periods of recruitment and data collection.</i>
Outcomes	<i>Describe how you pre-defined primary and secondary outcome measures and how you assessed these measures.</i>

Dual use research of concern

Policy information about [dual use research of concern](#)

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes	
☐	☐	Public health
☐	☐	National security
☐	☐	Crops and/or livestock
☐	☐	Ecosystems
☐	☐	Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No Yes

- | | | |
|--------------------------|--------------------------|---|
| ☐ | ☐ | Demonstrate how to render a vaccine ineffective |
| ☐ | ☐ | Confer resistance to therapeutically useful antibiotics or antiviral agents |
| ☐ | ☐ | Enhance the virulence of a pathogen or render a nonpathogen virulent |
| ☐ | ☐ | Increase transmissibility of a pathogen |
| ☐ | ☐ | Alter the host range of a pathogen |
| ☐ | ☐ | Enable evasion of diagnostic/detection modalities |
| ☐ | ☐ | Enable the weaponization of a biological agent or toxin |
| ☐ | ☐ | Any other potentially harmful combination of experiments and agents |

Plants

Seed stocks

n/a

Novel plant genotypes

n/a

Authentication

n/a

ChIP-seq

Data deposition

- ☐ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links

May remain private before publication.

For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.

Files in database submission

Provide a list of all files available in the database submission.

Genome browser session
(e.g. [UCSC](#))

Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.

Methodology

Replicates

Describe the experimental replicates, specifying number, type and replicate agreement.

Sequencing depth

Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.

Antibodies

Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.

Peak calling parameters

Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.

Data quality

Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.

Software

Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.

Flow Cytometry

Plots

Confirm that:

- ☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☐ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.

Instrument

Identify the instrument used for data collection, specifying make and model number.

Software

Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.

Cell population abundance

Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.

Gating strategy

Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.

- ☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type

Indicate task or resting state; event-related or block design.

Design specifications

Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.

Behavioral performance measures

State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).

Acquisition

Imaging type(s)

Specify: functional, structural, diffusion, perfusion.

Field strength

Specify in Tesla

Sequence & imaging parameters

Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.

Area of acquisition

State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.

Diffusion MRI

☐ Used

☐ Not used

Preprocessing

Preprocessing software

Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).

Normalization

If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.

Normalization template

Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.

Noise and artifact removal

Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).

Volume censoring

Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.

Statistical modeling & inference

Model type and settings

Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).

Effect(s) tested

Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.

(See [Eklund et al. 2016](#))

Correction

Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).

Models & analysis

n/a | Involved in the study

- ☐ ☐ Functional and/or effective connectivity
- ☐ ☐ Graph analysis
- ☐ ☐ Multivariate modeling or predictive analysis

Functional and/or effective connectivity

Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).

Graph analysis

Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).

Multivariate modeling and predictive analysis

Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.